



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE (Main)

UNIT TEST

09-05-2021

JEE(Main + Advance) : Enthusiast Course (PHASE : S, I(A), I, LIVE-I(A) & LIVE-I)

TEST DATE : 09-05-2021

ANSWER KEY

PAPER-2

PHYSICS

SEC-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	B	C	A	B	A	C	A	C	B
SEC-B	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	C	A	D	C	B	C	C	C	D
SEC-B	Q.	1	2	3	4	5	6	7	8	9	10
	A.	100.00	5.00	3.50	12.50	50.00	1.00	24.00	60.00	1.33	3.00

CHEMISTRY

SEC-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	B	A	D	B	D	D	C	A	B
SEC-B	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	B	C	C	D	B	B	B	D	A
SEC-B	Q.	1	2	3	4	5	6	7	8	9	10
	A.	9.00	3.00	5.00	4.00	4.00	3.00	2.00	6.00	7.00	1.20

MATHEMATICS

SEC-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	C	A	B	A	C	D	A	C	A
SEC-B	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	D	C	C	B	B	A	A	B	D
SEC-B	Q.	1	2	3	4	5	6	7	8	9	10
	A.	4.00	0.00	3.00	0.00	4.00	1.00	5.00	1.00	1.00	0.42 or 0.43

TEST DATE : 09-05-2021
PAPER-2
HINT – SHEET
PHYSICS
SEC-A
1. Ans (B)

$$\frac{\omega_A}{f_A} + \frac{\omega_B}{f_B} = 0$$

$$\Rightarrow \frac{f_A}{f_B} = -\frac{\omega_A}{\omega_B} = -\frac{2}{3}$$

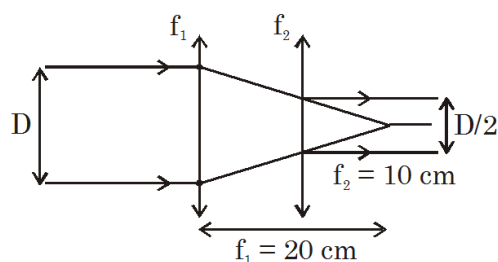
$$\Rightarrow f_A = -\frac{2}{3}f_B$$

$$\Rightarrow \frac{1}{f_A} + \frac{1}{f_B} = P_{eq} = -0.8D$$

$$-\frac{3}{2f_B} + \frac{1}{f_B} = -\frac{0.8}{100}$$

$$-\frac{1}{2f_B} = -\frac{0.8}{100}$$

$$f_B = \frac{100}{1.6} \text{ cm} = 62.5 \text{ cm}$$

2. Ans (B)


$$x = 20 - 10 = 10 \text{ cm}$$

3. Ans (C)

$$E = \frac{-dV}{dr}$$

$$\int dV = |\vec{E} \cdot d\vec{r}| = \Delta V$$

 If $\vec{E}$ and $d\vec{r}$ are parallel and $\Delta V = 2$ volts

 $\Rightarrow |\vec{E}| = 20 \text{ V/m}$ but if $\vec{E}$ and $d\vec{r}$ are not parallel $|\vec{E}| > 20 \text{ V/m}$
4. Ans (A)

$$h_0 = \sqrt{h_1 h_2}$$

$$\Rightarrow R_0 = \sqrt{R_1 R_2} \quad (R \rightarrow \text{Radius})$$

$$\text{Area} \propto R^2$$

$$\therefore \text{Area of object} = \sqrt{(2.5)(10)} = 5 \text{ mm}^2$$

5. Ans (B)

$$v = \omega \sqrt{A^2 - \left(\frac{A}{3}\right)^2} = \frac{2\sqrt{2}A\omega}{3}$$

$$v_{\text{new}} = 2v = \frac{4\sqrt{2}A\omega}{3} = \omega \sqrt{(A_{\text{new}})^2 - x^2}$$

$$x = \frac{A}{3} \Rightarrow A_{\text{new}} = \frac{\sqrt{33}A}{3}$$

6. Ans (A)

$$y_1 = 6 \cos \left[6\pi t + \frac{\pi}{6} \right]$$

$$y_2 = 3(\sqrt{3} \sin 3\pi t + \cos 3\pi t)$$

$$= 3[2 \sin(3\pi t + \frac{\pi}{6})]$$

$$= 6 \sin[3\pi t + \frac{\pi}{6}]$$

$$A_1 : A_2 = 6 : 6 = 1 : 1$$

7. **Ans (C)**

For F to be along negative x-axis, q_1 has to be negative while q_2 has to be positive.

$$\text{Also } F_1 \cos 53^\circ = F_2 \cos 37^\circ$$

$$F_1 = \frac{kq_1q_3}{4^2}, F_2 = \frac{kq_2q_3}{3^2}$$

on putting value

$$q_2 = \frac{27}{32} \mu C$$

8. **Ans (A)**

$$k_1 x_1 = k_2 x_2 \Rightarrow \frac{x_1}{x_2} = \frac{k_2}{k_1} = \frac{1}{2}$$

$$\text{When block is at A, then } x_1 = \frac{A}{3}, x_2 = \frac{2A}{3}$$

$$E = \frac{1}{2} k_1 x_1^2 + \frac{1}{2} k_2 x_2^2$$

$$E = \frac{1}{2} (2k) \left(\frac{A}{3} \right)^2 + \frac{1}{2} (k) \left(\frac{2A}{3} \right)^2 = \frac{kA^2}{3}$$

When block is at $\frac{A}{2}$, extension of spring 1 is

$$\frac{A}{6}$$

$$\therefore E' = \frac{1}{2} (2k) \left(\frac{A}{6} \right)^2 = \frac{kA^2}{36} = \frac{1}{12} \left(\frac{kA^2}{3} \right)$$

$$= \frac{E}{12}$$

9. **Ans (C)**

$$\delta_1 + \delta_2 = 0$$

$$(\mu_1 - 1) A_1 - (\mu_2 - 1) A_2 = 0$$

10. **Ans (B)**

$$\text{Impulse 1} = 9 \times 0.02 = 0.18$$

$$\text{Impulse 2} = -9 \times 0.01 = -0.09$$

$$\text{Net Impulse} = 0.09 = \Delta P$$

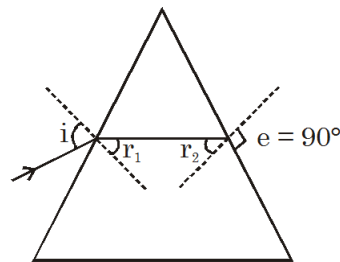
$$0.09 = m(\Delta v)$$

$$0.09 = 3(\Delta v)$$

$$\Delta v = 0.03 \text{ m/sec} = 3 \text{ cm/sec}$$

11. **Ans (A)**

For minimum i , e should be maximum and equal to 90°



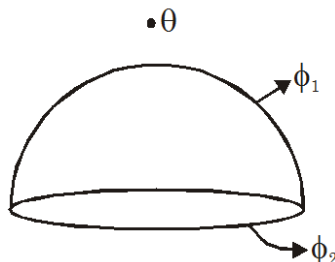
$$1. \sin i = \sqrt{\frac{7}{3}} \sin r_1$$

$$\sqrt{\frac{7}{3}} \sin r_2 = 1. \sin 90^\circ$$

$$\text{and } r_1 + r_2 = A = 60^\circ$$

on solving for i , we get $i = 30^\circ$

12. **Ans (C)**



$$\phi_1 + \phi_2 = 0$$

$$\text{Given flux through flat surface} = \phi_2 = \phi$$

$$\therefore \phi_1 = -\phi_2 = -\phi$$

13. **Ans (A)**

$$e = \frac{v_2 - v_1}{u_1 - u_2} = \frac{1 - 0.2}{0.8 - 0} = 1$$

14. **Ans (D)**

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$\frac{\mu_3}{v} - \frac{\mu_2}{v} = \frac{\mu_3 - \mu_2}{R}$$

$$\Rightarrow \frac{\mu_3}{v} - \frac{\mu_1}{u} = \frac{\mu_3 + \mu_1 - 2\mu_2}{R}$$

15. **Ans (C)**

$$\lambda (\pi R g) \left(\frac{2R}{\pi} \right) = (\lambda L g) \frac{L}{2}$$

$$\frac{L}{R} = 2$$

16. Ans (B)

$$y = 2t^2$$

$$a = \frac{d^2y}{dt^2} = 4$$

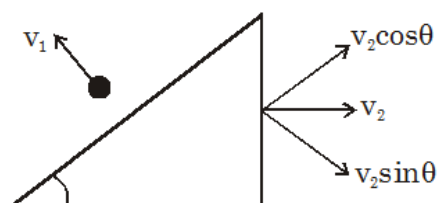
$$T = 2\pi\sqrt{\frac{\ell}{g_{\text{eff}}}}$$

$$T_1 = 2\pi\sqrt{\frac{\ell}{g}}$$

$$T_2 = 2\pi\sqrt{\frac{\ell}{g+a}}$$

$$\frac{T_2}{T_1} = \sqrt{\frac{g}{g+a}} = \sqrt{\frac{10}{14}} = \sqrt{\frac{5}{7}}$$

17. Ans (C)



$$e = \frac{v_2 - v_1}{u_1 - u_2}$$

$$\frac{7}{20} = \frac{v_1 + v_2/2}{10 - 0} \Rightarrow 2v_1 + v_2 = 7 \quad \dots (1)$$

From momentum conservation along horizontal.

$$(1) (1) \sin 30^\circ = -1 (v_1 \sin 30^\circ) + 2v_2$$

$$\Rightarrow 5 = 2v_2 - \frac{v_1}{2} \quad \dots (2)$$

From (i) and (ii)

$$v_1 = 2 \text{ m/s}$$

18. Ans (C)

$$\phi_{\text{net}} = \frac{q_{\text{in}}}{\epsilon_0}$$

19. Ans (C)

$$\vec{F} = -10\hat{i} + 5\hat{j}$$

$$\vec{F}_g = 10\hat{j}$$

$$\vec{F}_{\text{net}} = -10\hat{i} - 5\hat{j}$$

$$\vec{a} = -10\hat{i} - 5\hat{j}$$

for $y = 0$

$$0 = 10t - \frac{1}{2} \times 10t^2 = 0$$

$$t = 0, t = 2$$

at $t = 0$, $x = 0$, & at $t = 2$,

$$x = 0 + \frac{1}{2} (5) (2)^2 = 10$$

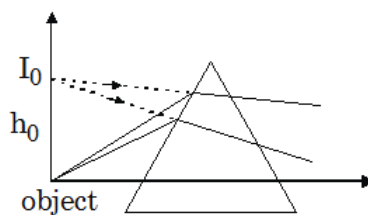
$\therefore$ it passes through origin.

at origin, $v_x = -10\hat{i}$

$$v_y = -10\hat{j}$$

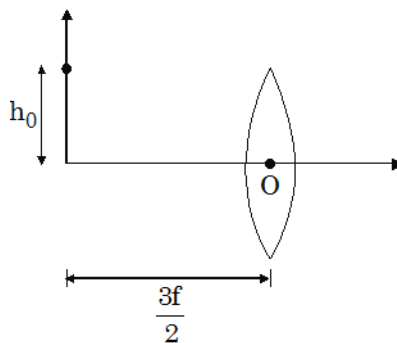
$$v_{\text{net}} = 10\sqrt{2}$$

20. Ans (D)



$$\delta = (n - 1) \times \alpha$$

$$h_0 = f(n - 1) \alpha$$



$$\frac{1}{v} - \frac{2}{(-3f)} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{f} - \frac{2}{3f}$$

$$\frac{1}{v} = \frac{3 - 2}{3f} \Rightarrow v = 3f$$

$$\frac{h_2}{h_1} = \frac{v}{u} \Rightarrow h_2 = h_0 \frac{3f}{-3f/2}$$

$$h_2 = -2f(n - 1)\alpha$$

PHYSICS

SEC-B

1. Ans (100.00)

$$\vec{E} = -\frac{\partial v}{\partial x} \hat{i} - \frac{\partial v}{\partial y} \hat{j} = -3\hat{i} - 4\hat{j}$$

$$|\vec{E}| = 5 \text{ V/m}$$

$$\frac{U}{\text{vol}} = \frac{1}{2} \epsilon_0 E^2$$

$$U = \frac{1}{2} \epsilon_0 E^2 (2)^3 = \frac{1}{2} \epsilon_0 (5)^2 (2)^3$$

$$= 100.00$$

2. Ans (5.00)

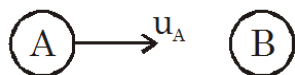
$$\ell = f_0 + f_e = 1$$

$$f_e = 20 \text{ cm}$$

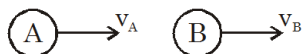
$$m = \frac{f_0}{f_e} = \frac{80}{20} = 4$$

$$m = \frac{f_e}{m} = \frac{20}{4} = 5.00$$

3. Ans (3.50)



After collision



$$P_i = P_f$$

$$mu_A = mv_A + 2mv_B$$

$$16 = v_A + 2v_B$$

Ball B reaches 5m height after collision

$$\therefore \frac{1}{2}(2m)v_B^2 = (2m)(g)(5)$$

$$v_B^2 = 100$$

$$\Rightarrow v_B = 10 \text{ m/s}$$

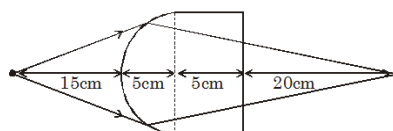
$$\Rightarrow v_A = -4 \text{ m/s}$$

$$e = \frac{v_B - v_A}{u_A - u_B} = \frac{10 - (-4)}{16 - 0}$$

$$e = \frac{14}{16}$$

$$ev_A = \frac{14}{16} \times 4 = 3.50$$

4. Ans (12.50)



For curved surface

$$\frac{1.6}{v_1} - \frac{1}{-15} = \frac{1.6 - 1}{5}$$

$$\frac{1.6}{v_1} = \frac{0.6}{5} - \frac{1}{15}$$

$$\frac{1.6}{v_1} = \frac{9 - 5}{75}$$

$$v_1 = \frac{75 \times 1.6}{4} = 30 \text{ cm}$$

For flat surface

$$\frac{1}{v_2} - \frac{1.6}{20} = 0$$

$$\Rightarrow v_2 = \frac{20}{1.6} = \frac{200}{16} = 12.50 \text{ cm}$$

5. Ans (50.00)

$$-\int_0^{\Delta t} T dt = (1.25)v_f - (0.25) \times 400 \text{ and}$$

$$\int_0^{\Delta t} T dt = (0.75)v_f - 0 \Rightarrow 2v_f = \frac{1}{4} \times 400$$

$$\text{HS-4/8} \Rightarrow v_f = 50 \text{ m/s}$$

6. Ans (1.00)

$$\int_{x=0}^{x=\infty} E dx = v_0 - v_{\infty}$$

$$= \frac{kQ}{R} = \frac{9 \times 10^9 \times 0.1 \times 10^{-9}}{0.9} = 1.00$$

7. Ans (24.00)

$$\frac{1}{f_1} = \left(\frac{\mu_1}{\mu_{\omega}} - 1 \right) \left(\frac{1}{R} - \left(-\frac{1}{R} \right) \right)$$

$$\frac{1}{f_1} = \left(\frac{\mu_1}{\mu_{\omega}} - 1 \right) \left(\frac{2}{R} \right)$$

Similarly

$$\frac{1}{f_2} = \left(\frac{\mu_2}{\mu_{\omega}} - 1 \right) \left(-\frac{2}{R} \right)$$

$$\frac{1}{f_{eq}} = \frac{1}{f_1} + \frac{1}{f_2} = \left(\frac{2}{R} \right) \left[\frac{\mu_1}{\mu_{\omega}} - 1 - \frac{\mu_2}{\mu_{\omega}} + 1 \right]$$

$$\frac{1}{f_{eq}} = \frac{2}{R} \left[\frac{\mu_1 - \mu_2}{\mu_{\omega}} \right]$$

$$\frac{1}{f_{eq}} = \frac{2}{20} \times \frac{5}{9} \times \frac{3}{4}$$

$$f_{eq} = 24 \text{ cm}$$

8. Ans (60.00)

For concave mirror

$$u = -120 \text{ cm}$$

$$f = -40 \text{ cm}$$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

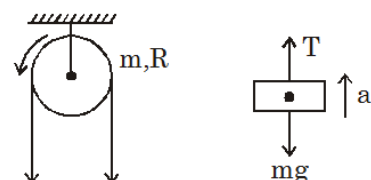
$$v = 60 \text{ cm}$$

9. Ans (1.33)

For disc, from torque equation

$$3mgR - TR = \frac{mR^2}{2}\alpha \quad \dots(1)$$

By application of Newton's second law on block we get,



$$T - mg = ma \quad \dots(2)$$

$$\text{where } a = R\alpha \quad \dots(3)$$

$$\text{solving } a = \frac{4g}{3}$$

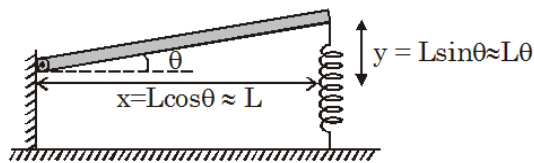
$$\therefore x = 0.133$$

10. Ans (3.00)

Restoring torque : $\tau = ky L = KL^2\theta$

(Since $y \cong L\theta$ from figure)

$$\Rightarrow kL^2\theta = \frac{mL^2}{3} \cdot \alpha \Rightarrow \alpha = \frac{3k}{m} \cdot \theta$$



$$\Rightarrow \omega = \sqrt{\frac{3k}{m}} \quad (\text{torque due to } mg \text{ was already}$$

balanced so it is not taken in calculation)

CHEMISTRY

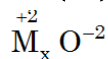
SEC-A

4. Ans (D)

In absence of '2s' and '2p' orbitals mixing, the MOT configuration of C_2 would be; $\sigma 1s^2 \sigma 1s^2 \sigma 2s^2 \sigma 2s^2 \sigma 2p_z^2 p 2p_x^1 = p 2p_y^1$. Hence it contains two unpaired electron and so it is paramagnetic.

Bond order = .

6. Ans (C)



Let $M^{2+} \rightarrow a$, $M^{3+} \rightarrow x - a$

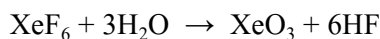
$$a \times 2 + (x - a) \times 3 = 2$$

$$\Rightarrow a = 3x - 2$$

$$x - a = 2 - 2x$$

$$\frac{M^{3+}}{M^{2+}} = \frac{2(1-x)}{3x-2}$$

13. Ans (C)



18. Ans (B)

$$\frac{P^0 - P_s}{P^0} = \frac{w_1 \times m_2}{m_1 \times w_2} \Rightarrow \frac{P^0 - P_s}{P^0} = \frac{(w_1 \times 1000)}{(m_1 \times w_2)} \times$$

$$\frac{m_2}{1000} = m \times \frac{m_2}{1000}$$

$$\left(\begin{array}{l} m = \text{molality} \\ m_2 = \text{mol wt. of solvent} \end{array} \right)$$

$$\Rightarrow \frac{760 - 750}{750} = m \times \frac{18}{1000}$$

$$\therefore m = \frac{10 \times 1000}{750 \times 18} = 0.741 \text{ mol/kg of solvent}$$

CHEMISTRY

SEC-B

4. Ans (4.00)

$$r \propto \frac{1}{\sqrt{M}} \Rightarrow \frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}}$$

$$\Rightarrow \frac{V}{t_1} \times \frac{t_2}{V} = \sqrt{\frac{M_2}{M_1}} \quad \left\{ \begin{array}{l} t_1 = 2 \text{ min} \\ t_2 = 5.65 \text{ min} \\ M_2 = 32 \end{array} \right.$$

$$\Rightarrow \frac{5.65}{2} = \sqrt{\frac{32}{M_1}}$$

$$\Rightarrow \frac{32}{M_1}$$

$$\Rightarrow M_1 = 4$$

MATHEMATICS

SEC-A

1. Ans (B)

$$y = \frac{x}{(1+x^2)}$$

$$x^2 y + y = x$$

$$x^2 y - x + y = 0$$

$$D \geq 0 \Rightarrow 1 - 4y^2 \geq 0$$

$$-\frac{1}{2} \leq y \leq \frac{1}{2}$$

2. Ans (C)

$f(x)$ is defined when $x \neq \pm 2$ and $x^3 - x > 0$

$$\Rightarrow x \neq \pm 2 \text{ and } x(x^2 - 1) > 0$$

$$\Rightarrow x \neq \pm 2 \text{ and } x \in (-1, 0) \cup (1, \infty)$$

3. Ans (A)

$$\lim_{x \rightarrow 1} \left[\sec\left(\frac{\pi x}{2}\right) \log x \right]$$

$$= \lim_{x \rightarrow 1} \frac{\log x}{\cos\left(\frac{\pi x}{2}\right)} = \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 1} \frac{\frac{1}{x}}{-\sin\left(\frac{\pi x}{2}\right) \cdot \frac{\pi}{2}} = -\frac{1}{\frac{\pi}{2} \sin \frac{\pi}{2}} = -\frac{2}{\pi}$$

4. Ans (B)

By L-Hospital rule

5. Ans (A)

Do Yourself.

6. **Ans (C)**

Definition of $[x]$ and L-Hospital rule.

7. **Ans (D)**

$$\text{Put } x = \frac{1}{2}, \text{ G.E.} = \frac{\pi}{6}$$

8. **Ans (A)**

$$\text{Put } x_1 = x_2 = x_3 = \dots = 1$$

$$\sum_{k=1}^n x_k = 1 + 1 + 1 + \dots + 1 \text{ (n times)} = n$$

9. **Ans (C)**

$$\text{Given } 3x - x^2 \geq 2 \Rightarrow x^2 - 3x + 2 \leq 0 \Rightarrow x = 1(a)2$$

At $x = 2$ sine is not defined

So $\sin^{-1}(1) + \sin^{-1}(1) + \dots + 10 \text{ times}$

$$= \frac{\pi}{2} + \frac{\pi}{2} + \dots + 10 \text{ times} = 5\pi$$

10. **Ans (A)**

$$\lim_{x \rightarrow 0} \frac{\cos ax - \cos bx}{x^2} = \frac{b^2 - a^2}{2} \quad \text{and}$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

11. **Ans (D)**

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \left(\frac{e^x - 1}{x} \right)^{2n} \cdot \frac{(x/a)^n}{\sin^n(x/a)} \times \frac{(x/a)^n}{(\log(1 + (x/a)))^n} \cdot a^{2n}$$

$$= a^{2n}$$

$$\text{since } f(0) = \lim_{x \rightarrow 0} f(x) \text{ so}$$

$$a^{2n} = 16^n = 4^{2n} \text{ thus } a = 4$$

12. **Ans (D)**

$$f(x) = x^{3/2} - \sqrt{x^3 + x^2}$$

$f(x)$ is differentiable & continuous at $x = 2, 5, 1$

$$\text{at } x = 0 \Rightarrow f(0) = f(0+h) = f(0-h)$$

$$f'(0^+) = \text{R.H.D.} = \frac{f(0+h) - f(0)}{h} = \sqrt{h} - \sqrt{h+1}$$

$$= -1$$

13. **Ans (C)**

$$f'(x) = \frac{xe^{-x^2}}{\sqrt{1 - e^{-x^2}}} \text{ is not differentiable only at}$$

$$x = 0$$

14. **Ans (C)**

$$f'(0^+) = \lim_{x \rightarrow 0} \frac{x^a - 0}{x - 0} = \lim_{x \rightarrow 0} \frac{1}{x^{1-a}}$$

$0 < a < 1 \Rightarrow (1-a) > 0 \Rightarrow f(0) = \infty$, does not exist

15. **Ans (B)**

$$g'(x) = f^{(ex)} \cdot e^{f(x)} \cdot f'(x) + e^{f(x)} \cdot f'(e^x) \cdot e^x$$

$$\therefore g'(0) = 8$$

16. **Ans (B)**

x-coordinates of foci are equal

$$|SP - S^1P| = 2b = 8 \Rightarrow b = 4, SS^1 = 2be$$

$$a^2 = b^2 e^2 - b^2, \text{ centre is midpoint of } S \text{ \& } S$$

17. **Ans (A)**

The given ellipse is

$$\Rightarrow a = 2, b = \sqrt{3} \Rightarrow 3 = 4(1 - e^2) \Rightarrow e = \frac{1}{2}$$

So that $ae = 1$

Hence the eccentricity e_1 of the hyperbola is

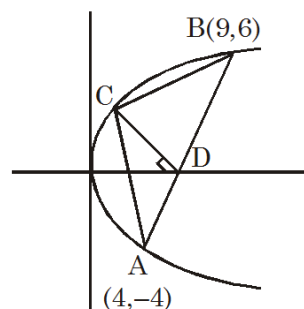
$$\text{given by } 1 = e_1 \sin \theta \Rightarrow e_1 = \text{cosec } \theta$$

$$\Rightarrow b^2 = \sin^2 \theta (\text{cosec}^2 \theta - 1) = \cos^2 \theta$$

$$\text{Hence the hyperbola is } \frac{x^2}{\sin^2 \theta} - \frac{y^2}{\cos^2 \theta} = 1 \text{ or}$$

$$x^2 \text{cosec}^2 \theta - y^2 \sec^2 \theta = 1$$

18. **Ans (A)**



Area of triangle ABC is maximum if CD is maximum tangent drawn to parabola at C should be parallel to AB.

$$\text{slope of AB} = \frac{6+4}{9-4} = 2$$

$$y^2 = 4x \Rightarrow \frac{dy}{dx} = \frac{2}{y} = 2 \Rightarrow y = 1$$

$$\therefore x = \frac{1}{4}$$

$$\text{Point is } \left(\frac{1}{4}, 1 \right)$$

19. Ans (B)

$$A = (a, 0), A^1 = (-a, 0)$$

$$\text{Tangent at } 6\theta, \frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1$$

$$x = \pm a \text{ at } L \text{ and } M$$

$$L = \left(a, \frac{b(1 - \cos \theta)}{\sin \theta}\right), M = \left(-a, \frac{b(1 + \cos \theta)}{\sin \theta}\right)$$

$$\Rightarrow AL \cdot A^1M = b^2$$

20. Ans (D)

$$a = 5, b = 4$$

$$(A) \frac{2b^2}{a} = \frac{32}{5}$$

$$(B) b^2 = a^2(e^2 - 1)$$

$$e^2 = \frac{41}{25} \Rightarrow e = \frac{\sqrt{41}}{5}$$

$$(C) \text{ Centre } (0, 0)$$

$$(D) 2ae = 2.5 \cdot \frac{\sqrt{41}}{5} = 2\sqrt{41}$$

MATHEMATICS

SEC-B

1. Ans (4.00)

$$\frac{3^2 + (3^2)^2 + (3)^3 + \dots + (3^n)^2}{\left(\frac{1}{3}\right)^2 + \left(\frac{1}{3^2}\right)^2 + \dots + \left(\frac{1}{3^n}\right)^2} = 3^{10}$$

$$3^{2n+2} = 3^{10}$$

$$n = 4$$

2. Ans (0.00)

Do Yourself.

3. Ans (3.00)

$$\text{Let the hyperbola be } \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \text{ then}$$

$$2a = ae, \text{ i.e., } e = 2$$

$$\therefore \frac{b^2}{a^2} = e^2 - 1 = 3 \quad \therefore \frac{(2b)^2}{(2a)^2} = 3$$

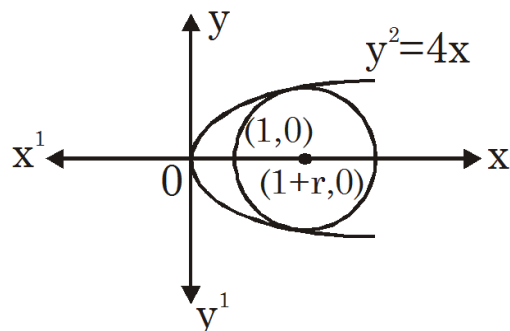
4. Ans (0.00)

$$\text{Perpendicular slope } m = \frac{2}{5}, \sqrt{a^2 m^2 - b^2} < 0$$

5. Ans (4.00)

Let r be the radius of the largest circle passing through the focus $(1, 0)$ of $y^2 = 4x$.

Clearly, centre of the circle will be on x -axis its co-ordinates are $(1 + r, 0)$



The equation of the circle is

$$(x - 1 - r)^2 + y^2 = r^2$$

It touches $y^2 = 4ax$. Therefore, the equation $(x - r - 1)^2 + 4x = r^2$ must have equal roots.

$$\Rightarrow 4(1 - r)^2 - 4(2r + 1) = 0 \Rightarrow r = 4$$

6. Ans (1.00)

$$16a^2 + 9(a - 1)^2 - 16a < 0$$

$$25a^2 - 34a + 9 < 0 \Rightarrow \frac{9}{25} < a < 1$$

7. Ans (5.00)

$$\Rightarrow x^3 - 2x - 1 = 0 \Rightarrow (x + 1)(x^2 - x - 1) = 0$$

$$\text{where } x = -1 \notin (\sqrt{2}, \sqrt{3})$$

$$\Rightarrow x^2 - x - 1 = 0 \Rightarrow x^4 = 3x + 2$$

$$\Rightarrow x^4 - \frac{3}{x} = (3x + 2) - (x^2 - 2)$$

8. Ans (1.00)

$$\text{Put } x = y = 1 \Rightarrow f(1) = 1$$

$$\text{put } y = 1 \Rightarrow f(x) = f\left(\frac{3}{x}\right), \forall x > 0$$

$$f(x) \cdot f\left(\frac{3}{x}\right) = 1 \Rightarrow f(x) = 1 \quad (\because x > 0)$$

9. Ans (1.00)

 At $x = -1$

$$\text{LHL} = \lim_{x \rightarrow -1^-} \frac{1 + \cos \pi x}{(x+1)^2} = \lim_{x \rightarrow 0^+} \frac{1 + \cos \pi(-1+h)}{\pi^2 h^2}$$

 (put $x+1 = h$)

$$\text{So, LHL} = \frac{1}{2} = -a + b = \text{RHL}$$

$$\Rightarrow b - a = \frac{1}{2} \quad \dots (i)$$

 At $x = 1$

$$\text{LHL} = a + b$$

$$\text{and } \lim_{x \rightarrow 1^+} \frac{x-1}{\ln \sqrt{x}} = \lim_{h \rightarrow 0^+} \frac{2h}{\ln(1+h)} = 2$$

 (Put $x = 1+h$)

$$\text{So, } a + b = 2 \quad \dots (ii)$$

from (i) & (ii)

$$b = \frac{5}{4} \quad \& \quad a = \frac{3}{4}$$

10. Ans (0.42 or 0.43)

$$g(x) = f^{-1}(x) \quad \left| \quad f(x) = (2x - 3\pi)^5 + \frac{4}{3}x + \cos x \right.$$

$$g(f(x)) = x \quad \left| \quad x = \frac{3\pi}{2} \right.$$

$$g'(f(x))f'(x) = 1 \quad \left| \quad f\left(\frac{3\pi}{2}\right) = 2\pi \right.$$